XAI-Driven Feature Optimization for Single-Cell Localization

Byunghyuk Youn†, Hyejin Shin†, Seongae Kang, Min Kim, *Student Member, IEEE*, Juyeop Kim*, and Ohyun Jo*, *Member, IEEE*

***Abstract*—This letter proposes a novel single-cell TDOA localization framework utilizing multi-beam transmission, which eliminates the dependency on multiple base stations. To mitigate environmental instability and high dimensionality, the proposed method introduces a dual-stage pipeline: first, similarity-based data refinement filters unstable environmental noise, and second, SHAP-guided feature selection optimizes input dimensionality. Field experiments in a realistic environment demonstrate that this approach maintains robust localization accuracy while achieving up to a 60% reduction in training overhead.**

***Index Terms*—Time Difference Of Arrival, Feature Selection, Wireless Localization, Deep Learning.**

I. Introduction

WIRELESS localization technology provides real-time spatial awareness for location-based services [1], [2]. TDOA-based positioning improves localization accuracy by exploiting relative timing differences among received signals. However, conventional approaches generally rely on multi-cell measurements, resulting in high signaling overhead and limited validation with real 5G user equipment (UE) [3], [4]. Moreover, practical TDOA features are high-dimensional and sensitive to channel and environmental variations.

To address this challenge, we introduce a single-cell localization framework that leverages multi-beam configurations to estimate user position via the TDOA across beams. Such intra-cell localization capability within a single cell can significantly reduce the burden on the User Equipment (UE). Unlike conventional TDOA-based positioning methods that require synchronization among multiple geographically separated base stations, the proposed framework estimates the UE position using downlink multi-beam transmissions from a single gNB.

Deep Learning (DL) based representation learning improves robustness to nonlinear distortions, noise, and outliers [5]. In particular, pretraining on large, diverse datasets followed by fine-tuning can substantially improve the reliability of TDOA estimates, which in turn translates into higher downstream localization accuracy.

†Byunghyuk Youn and †Hyejin Shin contributed equally to this work. Ohyun Jo and Juyeop Kim are the corresponding authors.

Byunghyuk Youn, Seongae Kang, Min Kim, and Ohyun Jo are with the Department of Computer Science, Chungbuk National University, Cheongju, Republic of Korea, e-mail : ybh0311, kseongae, kmin010287, ohyunjochungbuk.ac.kr

Hyejin Shin and Juyeop Kim are with the Department of Computer Science, Sookmyung Women's University, Seoul, Republic of Korea, e-mail : shj312, jykim@sookmyung.ac.kr

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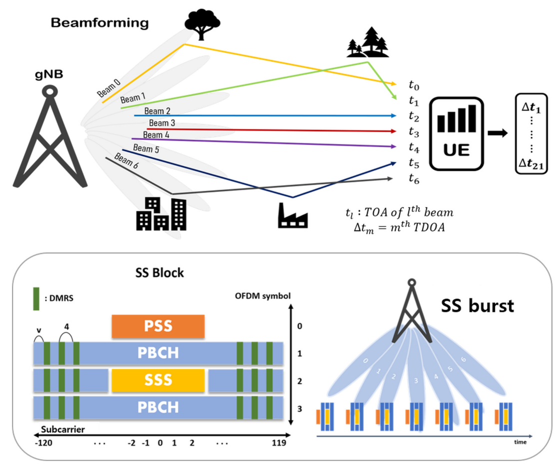


Fig. 1: Overview of beam-based TDOA measurement and the SS burst structure in 5G NR.

Explainable AI (XAI) offers a principled means to quantify per-feature contributions and to select features whose importance remains stable across scenarios [6]. Such XAI-based selection reduces input dimensionality to prevent overfitting and decreases latency and memory usage to satisfy real-time constraints. It also enhances generalization under distribution shifts and increases model transparency for improved trustworthiness [7]. Consequently, combining DL for nonlinear bias correction with XAI-based core feature retention provides a practical solution that enhances overall performance.

In this letter, we propose an XAI-driven wireless localization framework using single-cell TDOA data. The proposed method combines similarity-based data refinement and feature selection to improve data efficiency. Experimental results demonstrate its effectiveness under realistic environments.

II. System Models

Fig. 1 illustrates a beam-based TDOA measurement scenario in a 5G NR system, together with the structure of the transmitted synchronization signals based on SSBs. The gNobeB(gNB) periodically transmits a Synchronization Signal (SS) burst composed of L Synchronization Signal Blocks (SSBs), each of which is transmitted via a distinct beam. Each SSB contains the Primary SS (PSS), Secondary SS (SSS), and DeModulated Reference Signals (DMRSs), which are mainly used for synchronization and channel estimation purposes [8]. These signals are modulated using Orthogonal Frequency Division Multiplexing (OFDM), with the FFT size and subcarrier spacing denoted by N_{FFT} and f_{SCS}, respectively.

The UE estimates the integer and fractional parts of the Time of Arrival (TOA) of each beam based on the received

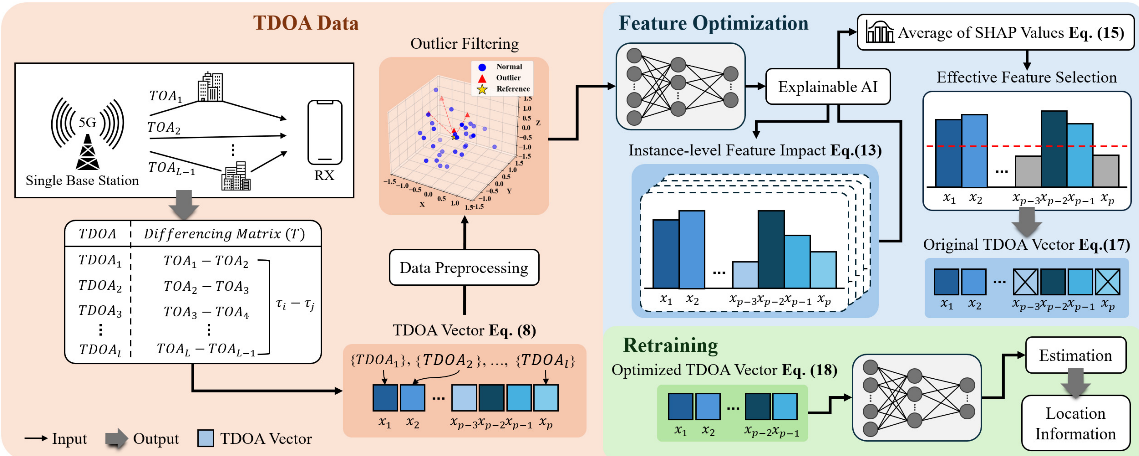


Fig. 2: Overall Framework of XAI-Guided Feature Selection for Model Optimization: The pipeline comprises three stages: (1) TDOA vector construction from single base-station measurements, (2) SHAP-based feature optimization, and (3) retraining with the compressed TDOA vector for accurate location estimation.

SSB. The Integer TOA (ITOA) of the l-th beam is obtained by correlating the received samples, denoted by $y[n]$, with the transmitted OFDM signal containing the PSS, denoted by $x_{PSS}[n]$. The estimated ITOA, denoted by $\hat{t}_l$, is derived as follows:

$$\hat{t}_l = \arg\max_{t \in \Delta_l} |R_l[t]|, \quad (1)$$

$$R_l[t] = \sum_{n=0}^{N_{\text{PSS}}-1} y[n+t] x^*_{PSS}[n]. \quad (2)$$

where N_{PSS} denotes the length of the transmitted PSS sequence, $x_{PSS}[n]$ denotes the transmitted PSS sequence, $(\cdot)^*$ denotes complex conjugation, and Δ_l represents the time window containing the received SSB corresponding to the l-th beam. Equation (2) provides a coarse TOA estimate with the resolution of one sampling period.

The UE then refines the RTOA of the l-th beam, denoted by ϵ_l, using the CFR derived from the received DMRS symbols. The phase difference of the CFR between two subcarriers spaced by M subcarrier intervals for the l-th beam is expressed as

$$\Delta\theta_l^{(M)} = -2\pi f_{\text{SCS}} M \epsilon_l. \quad (3)$$

In this case, we assume $M = 4$, since DMRS symbols are allocated every four subcarriers. Based on (3), ϵ_l can be derived as

$$\epsilon_l = -\frac{\Delta\theta_l^{(M)}}{2\pi f_{\text{SCS}} M}. \quad (4)$$

For noise suppression, the UE uses the averaged CFRs to estimate $\Delta\theta_l^{(M)}$.

For noise suppression, the UE estimates $\Delta\theta_l^{(M)}$ from the averaged CFRs at low- and high-frequency regions [9]. If we denote $\hat{H}_{\text{low},l}$ and $\hat{H}_{\text{high},l}$ as the averages of the estimated CFRs of the l-th beam over the low and high frequency subcarriers, respectively, (4) can be rewritten as:

$$\epsilon_l = -\frac{N_{\text{FFT}}}{2\pi M} \angle(\hat{H}_{\text{high},l} \hat{H}^*_{\text{low},l}). \quad (5)$$

The total TOA of the l-th beam is then obtained as

$$\tau_l = \hat{t}_l + \epsilon_l. \quad (6)$$

After estimating the TOA for all L beams, the UE constructs the estimated TOA vector denoted by $\boldsymbol{\tau}$:

$$\boldsymbol{\tau} = [\tau_1, \tau_2, \ldots, \tau_L]^T. \quad (7)$$

The TDOA feature vector representing inter-beam TOA differences is denoted by $\mathbf{x}$ and given by

$$\mathbf{x} = \mathbf{T}\boldsymbol{\tau}. \quad (8)$$

where $\mathbf{T} \in \mathbb{R}^{\binom{L}{2} \times L}$ is a differencing matrix that generates all possible inter-beam delay differences $(\tau_i - \tau_j)$ for $1 \le i < j \le L$. The TDOA feature vector serves as input to subsequent data-driven beam identification and localization algorithms.

III. XAI-Driven TDOA Feature Optimization

In practical wireless environments, multipath propagation caused by reflection, diffraction, and scattering, as well as measurement noise, can distort the residual delay ϵ_l differently across beams. These distortions are accumulated in the inter-beam delay differences represented by the TDOA feature vector $\mathbf{x} = \mathbf{T}\boldsymbol{\tau}$, thereby degrading feature reliability. Moreover, such channel-induced variations often produce complex and nonlinear feature distributions that are difficult to handle using simple rule-based criteria. Therefore, the proposed data-driven refinement and XAI-based feature selection are employed to mitigate unstable measurements and retain reliable TDOA features for localization.

A. Similarity-Based Data Refinement

TDOA measurements are susceptible to environmental variations and noise, which frequently introduce outliers that degrade predictive performance. To improve data quality, a similarity-based outlier detection method is applied to TDOA subsets collected from repeated experiments under the same coordinate framework. The cosine similarity between the average vectors of different subsets is computed as follows:

$$\mathrm{sim}(A_m, A_n) = \frac{A_m \cdot A_n}{\|A_m\| \, \|A_n\|}, \quad A_m \neq A_n \tag{9}$$

where A_m denotes the average vector of m-th subset S_m, A_n represents the average vector of n-th another subset S_n, and $\mathrm{sim}(A_m, A_n)$ indicates cosine similarity. Based on these similarities, a stability score is obtained by averaging the values across subsets:

$$\bar{A}_m = \sum_{A_m \in S,\, A_m \neq A_n} \mathrm{sim}(A_m, A_n) \tag{10}$$

where $\bar{A}_m$ represents the total similarity score of subset S_m. The final dataset $S = \{S_1, S_2, \ldots, S_M\}$ denotes the set of all subsets. M is the number of subsets. The refined sample set is then defined as follows:

$$\widetilde{S} = \left\{ S_i \;\middle|\; S_i \in S,\ \bar{A}_i \geq \delta \right\}. \tag{11}$$

Here, S_i represents the chosen subset out of S, $\widetilde{S}$ denotes the refined set of instances after filtering. δ is a stability threshold used to remove unstable subsets whose stability scores fall below this value. Cosine similarity focuses on directional consistency between TDOA feature distributions rather than absolute magnitude differences, making it suitable for identifying structurally inconsistent subsets.

B. XAI-Driven Feature Selection Strategy

As defined in the TDOA, values are obtained for all possible pairs of TOAs, and these values collectively form the input feature vector for model training. The prediction process is expressed as follows:

$$y = f(\mathbf{x}), \qquad \mathbf{x} = [x_1, \ldots, x_p] \in \mathbb{R}^p, \tag{12}$$

where f denotes the learning model, $\mathbf{x}$ is the same TDOA feature vector defined in Equation (8), y denotes the model prediction, and p is the number of TDOA features. Since a multi-beam scheme is employed, the dimension of the feature vector is given by $p = \frac{L(L-1)}{2}$.

we employ SHapley Additive exPlanations (SHAP) values as a model-interpretability-based feature selection method. SHAP values provide a quantitative measure of the importance of each input feature to the predictions of the model, derived from the trained model itself. The prediction is computed as the sum of all feature contributions. The method for calculating feature contributions is described as follows:

$$f(\mathbf{x}^i) = \phi_0 + \sum_{j=1}^{p} \phi_j(\mathbf{x}^i), \tag{13}$$

where $\mathbf{x}^i$ denotes an arbitrary feature vector within $\widetilde{S}$. $\phi_j(\mathbf{x}^i)$ represents the contribution of j-th feature to the prediction for sample $\mathbf{x}^i$. ϕ_0 denotes the model's baseline prediction in the absence of any feature information. The contribution of feature $\phi_j(\cdot)$ is derived via the following equation:

$$\phi_j(\mathbf{x}^i) = \sum_{\mathbf{x}_j^i \subseteq F \setminus \{j\}} \frac{|\mathbf{x}_j^i|!\,(p - |\mathbf{x}_j^i| - 1)!}{p!} \left(v(\mathbf{x}^i) - v(\mathbf{x}_j^i) \right), \tag{14}$$

where F denotes the full set of features, and $X_j^i \subseteq F \setminus \{j\}$ denotes a subset excluding feature j. $|\mathbf{x}_j^i|$ represents the cardinality of the subset, which indicates the number of features contained in the subset. The Shapley weight $\frac{|X_j^i|!\,(p-|X_j^i|-1)!}{p!}$ assigns uniform weighting over all $p!$ possible feature orderings. $v(\mathbf{x}^i)$ denotes the model output when the full feature set is provided, whereas $v(\mathbf{x}_j^i)$ represents the conditional expected model output when only the subset $\mathbf{x}_j^i$ is given.

After extracting SHAP values for all features, the values are aggregated over the entire dataset to quantify each feature's relative importance using the mean absolute SHAP value:

$$I_j \;=\; \frac{1}{\omega} \sum_{i=1}^{k} \left| \phi_j(\mathbf{x}^i) \right|, \qquad j = 1, \ldots, p\ , \tag{15}$$

where $I_j(\cdot)$ denotes the mean importance score associated with the j-th feature. ω denotes the number of samples. The extracted importance scores $I_j(\cdot)$ are sorted in descending order to differentiate features with strong influence on the model from those with low influence. Accordingly, the ranking function is defined as follows:

$$I_{\pi(1)} \geq I_{\pi(2)} \geq \cdots \geq I_{\pi(p)}, \tag{16}$$

where $\pi(j)$ denotes a permutation of feature indices that sorts the importance scores in descending order. Applying the feature selection, we retain only the features that constitute the final input feature set as follows:

$$F_\lambda = \{\, j \,|\, j \in F,\, I_{\pi(j)} \geq \lambda \,\}, \tag{17}$$

where λ is a hyperparameter and F_λ represents the set of feature vectors constructed by selecting features with high contribution. After determining F_λ based on SHAP values, the model is retrained using only the selected feature subset, as follows:

$$\hat{y} = f(x_{F_\lambda}) \qquad x_{F_\lambda} = [x_j]\,,\, j \in F_\lambda, \tag{18}$$

where $\mathbf{x}_{F_\lambda}$ denotes vectors composed of the selected feature indices in F_λ, and $\hat{y}$ represents the predicted output using the optimized features. The proposed feature selection reduces data and computational complexity, thereby improving computational efficiency and generalization performance. An overview of the proposed framework is illustrated in Fig. 2.

TABLE I: Accuracy and Training Time Comparison of Models with Baseline and Selected k Values

Model	Metric	Baseline	$k=21$	$k=19$	$k=16$	$k=13$	$k=10$	$k=7$	$k=4$	$k=1$
CatBoost	Accuracy (%)	89.79	**90.32**	**90.44**	**90.19**	89.44	88.25	86.39	78.42	51.02
	Training Overhead (ms)	8.324	**7.814**	**6.945**	**6.644**	**6.315**	**5.001**	**4.599**	**4.390**	**4.104**
Decision Tree	Accuracy (%)	89.24	**89.68**	**89.87**	**89.73**	**89.38**	**89.28**	87.07	84.83	48.30
	Training Overhead (ms)	0.720	**0.669**	**0.678**	**0.678**	**0.528**	**0.527**	**0.365**	**0.361**	**0.308**
ExtraTrees	Accuracy (%)	93.39	**93.73**	**93.62**	**93.60**	**93.42**	93.24	92.45	90.11	46.81
	Training Overhead (ms)	2.932	**1.658**	**1.637**	**1.667**	**1.474**	**1.372**	**1.298**	**1.370**	**1.794**
LightGBM	Accuracy (%)	93.80	**94.21**	**94.17**	**94.17**	93.63	93.15	92.01	88.66	58.04
	Training Overhead (ms)	1.835	**1.816**	1.957	**1.756**	**1.666**	**1.564**	**1.371**	**1.351**	**1.349**
Random Forest	Accuracy (%)	93.88	**94.16**	**94.19**	**93.96**	93.68	93.23	92.10	89.58	46.81
	Training Overhead (ms)	13.404	**12.282**	**12.440**	**13.050**	**10.084**	**10.149**	**6.658**	**6.589**	**4.994**
XGBoost	Accuracy (%)	92.85	**93.36**	**93.39**	**93.12**	92.59	91.96	90.54	84.80	58.07
	Training Overhead (ms)	5.335	**2.068**	**1.955**	**1.733**	**1.551**	**1.372**	**1.267**	**1.074**	**0.955**
DNN	Accuracy (%)	87.47	**87.62**	86.95	86.02	86.12	87.03	87.33	77.26	39.80
	Training Overhead (ms)	57.516	**52.615**	**52.320**	**51.719**	**57.162**	**51.772**	**52.490**	**51.663**	**45.453**

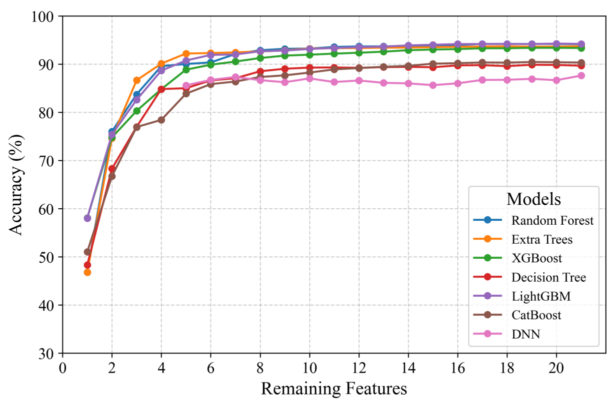
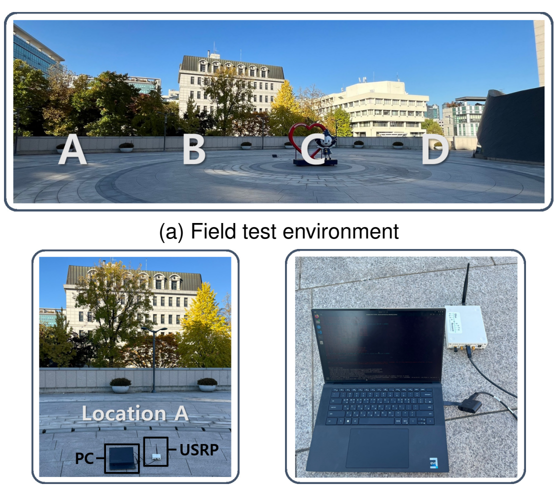


Fig. 3: Comparison of silhouette coefficients obtained from the baseline, proposed feature selection, and random feature selection models.



(a) Field test environment

(b) Measurement setup (c) Softwarized modem

Fig. 4: Field experiment environment and implementation.

IV. Experiment and Result

This section evaluates the proposed method by comparing the selected feature subset with the full feature baseline. In addition, clustering characteristics and structural separability in the reduced feature space are analyzed. The proposed approach is also compared with random feature selection to demonstrate the limitations of unstructured feature subsets.

A. Experimental Setup

To validate the practical viability of the proposed method, outdoor field experiments were conducted in a commercial wireless environment. Fig. 4 presents the experimental environment and implementation, including the test site, measurement setup using a commercial LGU+ gNB and USRP B210 receiver, and the Linux-based softwarized modem.

The system operated with $N_{\text{FFT}} = 1024$, $f_{\text{SCS}} = 30$ kHz, and $L = 7$ transmitted SSBs at a center frequency of 3.45936 GHz. TDOA feature vectors were repeatedly collected at four receiver locations shown in Fig. 4a, and localization accuracy was evaluated using single-cell feature vectors. In this work, localization performance was evaluated in terms of classification accuracy, where each receiver location was defined as an independent class.

Cosine similarity analysis was conducted across 11 dataset subsets, where subsets with average similarity scores lower than the stability threshold $\delta = 0.6$ were excluded to improve data reliability. All models were trained and evaluated using the identical refined dataset for fair comparison. To address model-dependent SHAP score distributions, a rank-based feature selection strategy was adopted instead of a fixed threshold λ. The top-k features were selected by progressively reducing k from 21 to 1 to analyze the impact of feature dimensionality. The SHAP-guided optimization was performed as an offline preprocessing step prior to model training.

B. Efficacy of Data Refinement and Feature Optimization

Table I presents the experimental results for various models. The refined dataset consistently improved localization performance, indicating that unstable subsets negatively affected training stability. In addition, stable performance was maintained even with progressively fewer selected features. In several cases, reducing approximately 10–25% of the features slightly improved performance, while using only about 50% of the features preserved comparable accuracy with up to 60% lower training overhead.

These results indicate that redundant features may hinder generalization, whereas selecting highly informative features improves learning robustness and computational efficiency.

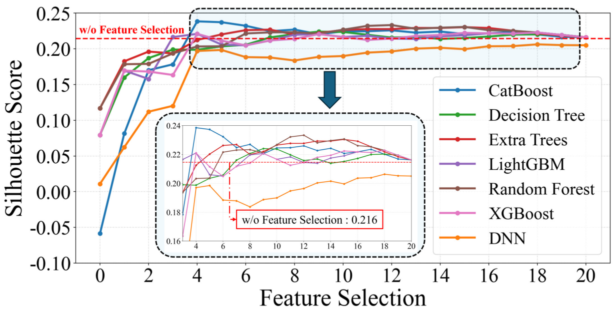


Fig. 5: Silhouette score comparison among models under different feature selection

C. Structural Consistency Analysis

Fig. 5 presents a comparative analysis of silhouette coefficients across various classification models as the number of selected features varies. The silhouette coefficient is a metric that accounts for both intra-cluster cohesion and inter-cluster separation. When the SHAP-based feature selection is applied, a general trend of improved silhouette scores is observed compared to using all features. In particular, models such as Random Forest and CatBoost achieve silhouette score improvements of up to 10%. This improvement arises because the SHAP-based feature selection removes irrelevant or redundant features that do not contribute to cluster formation while retaining features that play a critical role in data separation. As a result, the refined feature subsets preserve the structural consistency of the data distribution while more effectively enhancing inter-class separability. Furthermore, the consistent observation of similar trends across a wide range of classification models indicates that the proposed SHAP-based feature selection strategy is model-agnostic and robust in preserving the underlying cluster structure. These results indicate that the proposed feature selection strategy not only reduces feature redundancy but also effectively preserves the intrinsic geometric relationships among samples within the feature space.

D. Necessity of Importance-Aware Selection

Fig. 6 presents the silhouette scores obtained when features are selected at random. In contrast to Fig. 4, random feature selection consistently results in lower silhouette scores across all models. This degradation is not caused merely by the reduction in feature dimensionality, but rather by the removal of highly informative features that are essential for effective classification. Arbitrary feature selection eliminates key features that define cluster boundaries, thereby disrupting the structural consistency of the data distribution. In particular, as shown in Fig. 6 the removal of important features reduces the geometric separation between clusters and increases overlap among classes, which consequently weakens intra-cluster cohesion. These results demonstrate that feature selection is not simply a dimensionality reduction process, but an important optimization procedure for preserving the structural characteristics of the data and the discriminative capability of the model.

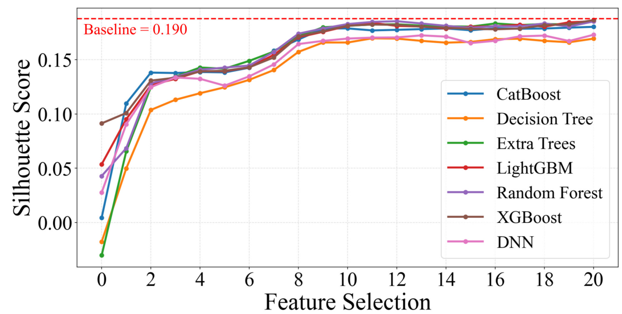


Fig. 6: Silhouette score comparison between models using random feature selection and those using all features

V. Conclusion

This letter presents an XAI-driven localization framework for single-cell 5G positioning. By leveraging SHAP-based feature optimization, the proposed method reduces training overhead by up to 60% while maintaining localization accuracy. Experimental Results demonstrate efficient localization through complexity reduction of high-dimensional TDOA features. Future work will investigate multi-antenna UE configurations and robust integration of antenna-wise TOA and TDOA features under diverse frequency bands and propagation environments.

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